

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

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## **EC-6001-CBGS**

### **B.E. VI Semester**

Examination, June 2020

## **Choice Based Grading System (CBGS)**

### **Cellular Mobile Communication**

*Time : Three Hours*

*Maximum Marks : 70*

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) With the help of block diagram describe the basic cellular system and discuss its performance criteria. 7  
b) Define frequency reuse. Why it is useful in cellular telephone system? Also discuss the problem arising due to frequency reuse. 7
2. a) Define and explain hand off mechanism. What is the difference between soft and hard hand off. 7  
b) Discuss the propagation over water or flat open area between fixed stations. 7
3. a) Explain point to point predictions model for obtaining path loss in obstructive condition. 7  
b) Describe antennas used at cell site. Discuss unique situations of cell site antenna. 7
4. a) Discuss about the real time co-channel interference measurement at mobile radio transceivers. 7  
b) What is umbrella-pattern effect? List the benefits of an umbrella cellular pattern. 7

EC-6001-CBGS

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[2]

5. a) What are the effect of coverage and interference by applying power decrease? 7  
b) What is non-fixed channel assignment? Explain different types of non-fixed channel assignment algorithms. 7
6. a) What is hand off mechanism explain: 10  
i) Initiation of hand off  
ii) Delaying a hand off  
iii) Forced hand off  
iv) Queuing of hand off  
b) Briefly explain the concept of frequency management. 4
7. a) Draw and explain and architecture of GSM. Also explain layer modeling. 7  
b) Describe Code Division Multiple Accessing (CDMA). Also discuss output power limit and control in CDMA. 7
8. Write short notes on (any two): 14  
a) Cell Splitting  
b) Mobile Antennas  
c) TDD System  
d) Mobile to Mobile Propagation

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